



End Semester Examination
Summer 2024

Max marks: 60

Duration: 2 hours

Branch: Computer Engineering

Semester: VI

Name of the Course: System Programming & Compiler Construction

Course code: CSC601

QP Code: R23-CSC601_012023-24

- N.B. (1) Attempt any three out of the five questions.
(2) Figures to the right indicate full marks.
(3) Assume suitable data if necessary.

	Marks
Q.1 (a) Explain the working of two pass macro processor with suitable example.	10
(b) What are the goals of system software? What are the functions of Loader and Linker?	10
Q.2 (a) What are the phases of compiler? Give the workings of each phase for the following statement- $a = a * b + c + e$ Variables a,b,c,d,e are of integer type.	10
(b) Explain with example i) Left factoring ii) Left Recursion	10
Write the steps for computing FIRST & FOLLOW	
Find First and Follow for the following grammar:	
$S \rightarrow ABC$	$\{a, c, f, d, e, b\}$ $\{b\}$
$A \rightarrow a \underline{C}b \epsilon$	$\{a, c, f, e\}$ $\{c, d, e, f, b\}$
$B \rightarrow c dA \epsilon$	$\{c, d, e\}$ $\{e, f\}$
$C \rightarrow c f$	$\{c, f\}$ $\{b, d\}$

QP Code: R23-CSC601_012023-24



- Q.3 (a) Explain features and types of macros? Explain Macro call and Macro expansion with suitable example. 10
- (b) Explain types of errors handled by pass I and pass II of the assembler. With reference to assembler explain the following tables with suitable example i) MOT ii) POT iii) ST iv) LT 10
- Q.4 (a) Explain the role of code optimization in compiler design. Explain machine independent code optimization techniques. 10
- (b) For the following grammar construct SLR parsing table. Show stack content, input buffer and action taken for any acceptable string. 10
- $S \rightarrow iCtSS' \mid a$
 $S' \rightarrow eS \mid \epsilon$
 $C \rightarrow b$
- Q.5 (a) Explain different storage allocation strategies in detail. 10
- (b) Explain with example Quadruples, Triples and Indirect Triples 10

$X \longrightarrow X \longrightarrow X \longrightarrow X$



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Branch: Computer Engineering

Name of the Course: Cryptography and System Security

Course code: CSC602

Duration: 2 hours

Semester: VI

QP Code: R23-CSC602_012023-24

- N.B. (1) Attempt any three out of the five questions.
(2) Figures to the right indicate full marks.
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Marks

- Q.1 (a) Compare and contrast Vigenère cipher and the Playfair cipher with merits and demerits. Show the encryption and decryption process of both for plaintext "it is a secret" with key "KEY". 10
(b) Describe the general structure of DES and give details of one round in DES. List out the weaknesses of DES in terms of design and cipher key. 10
- Q.2 (a) Demonstrate with help of algorithm and an example, how Diffie Hellman works as a key distribution technique. 10
(b) Compare various challenge-response methods for user authentication. 10
- Q.3 (a) Demonstrate how PGP achieves message integrity and confidentiality in email. 10
(b) Discuss (a)SQL Injection (b) Keyloggers (c)Spywares (d)ARPspoofing 10
- Q.4 (a) State Euclidean Algorithm in pseudo code. Demonstrate the working of the algorithm with an example. 10
(b) Define the properties and applications of hash functions. 5
(c) Find A's private key in a RSA cryptosystem, if participant A uses two prime numbers $p = 13$ and $q = 17$ to generate her public and private keys and the public key of A is given as 35. 5
- Q.5 (a) Explain working of Kerberos. 10
(b) Describe flooding and DOS attacks against network security. 10

X—X—X—X

QP Code: R23-CSC602_012023-24



**End Semester Examination
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Max marks: 60

Branch: Computer Engineering

Name of the Course: Mobile Computing

Course code: CSC603

Duration: 2 hours

Semester: VI

QP Code: R23-CSC603_012023-24

- N.B.** (1) Attempt any three out of the five questions.
(2) Figures to the right indicate full marks.
(3) Assume suitable data if necessary justify your answers with neat diagram wherever necessary.

	Marks
Q.1 (a) Differentiate between Infrastructure Network and Ad-hoc Network	05
(b) Compare 2G 3G 4G and 5G Mobile Generations	05
(c) Explain UMTS in detail with its domain and Interface block diagram	10
Q.2 (a) Explain GSM Architecture in Detail	10
(b) Explain Tunneling and encapsulation in brief. What is Reverse tunneling?	10
Q.3 (a) Explain how packets are delivered from one node to Mobile Node	10
(b) What are various GSM Security Mechanisms? Explain WEP in brief.	10
Q.4 (a) Explain Hidden Station and Exposed station problem with the solution to avoid such issues.	10
(b) What do you mean by EDGE Computing, Explain in Details With suitable examples	10
Q.5 (a) Write a Short Note on a) Bluetooth b) M-Commerce	10
(b) Explain Various Handover Mechanism in GSM.	10

X—X—X—X

QP Code: R23-CSC603_012023-24

b) Diff. betⁿ supervised & unsupervised learning with example.



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Branch: Computer Engineering

Name of the Course: Artificial Intelligence

Course code: CSC604

Duration: 2 hours

Semester: VI

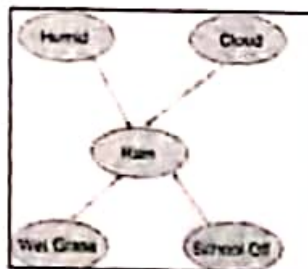
QP Code: R23-CSC604_02202324

- N.B.** (1) Attempt any three out of the five questions.
(2) Figures to the right indicate full marks.
(3) Assume suitable data if necessary

Marks

- Q.1 (a) Describe a) Ethics in AI b) PEAS Descriptors for the House cleaning robot. 5
(b) What are the problems that occur in Hill Climbing Technique? Illustrate with a neat diagram. 5
(c) Illustrate with an example the architecture of Utility Based Agent. 5
(d) $P(H)=0.2$ $P(W|R=T)=0.8$ $P(S|R=T)=0.7$
 $P(C)=0.6$ $P(W|R=F)=0.1$ $P(S|R=F)=0.2$ 5

$P(R|H=T, C=T)=0.9$ H=Humid C=Cloud S=School off
 $P(R|H=T, C=F)=0.4$ R=Rain W=Wet Grass T=True F=False
 $P(R|H=F, C=T)=0.7$
 $P(R|H=F, C=F)=0.1$



Find the probability of school being off when it is cloudy and raining but not humid.

- Q.3 (a) Explain Partial Order Planning with a suitable example

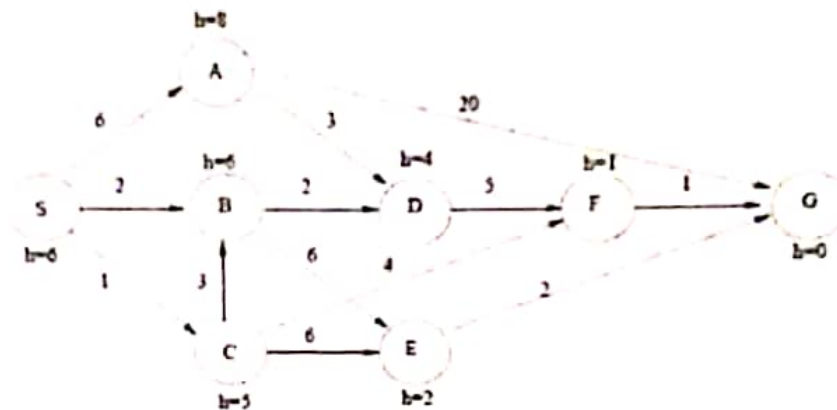
10

QP Code: R23-CSC604_02202324

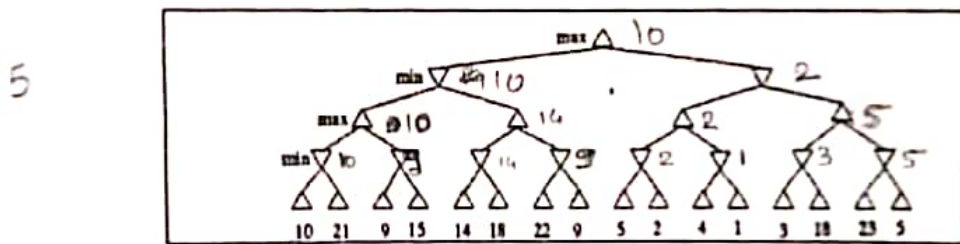


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- (b) Consider the graph given below. Assume that the initial state is S and the goal is G. Show how A* Search would create a search tree to find a path from the initial state to the goal state and find the cost of the solution and the optimal path. 10



- Q.4 (a) Apply alpha-beta pruning on the example given below with proper justifications. 10



- (b) Define the term Chromosomes, Fitness function, CrossOver and Mutation as used in Genetic algorithms. Explain how the Genetic algorithm works with help of the flowchart.
- Q.5 (a) 1. Sunil likes all kinds of food.
2. Apple and Vegetable are food
3. Anything anyone eats and is not killed is food
4. Anil eats peanuts and is still alive
5. Sohan eats everything that Anil eats

Prove by resolution that Sunil likes peanuts using resolution.

- (b) Explain Forward and Backward Chaining with a suitable example for each. 10

X-X-X-X-X-X

QP Code: R23-CSC604_02202324

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**End Semester Examination
Summer 2024**

Max marks: 60

Branch: Computer Engineering

Name of the Course: Quantitative Analysis

Course code: CSDLO6013

Duration: 2 hours

Semester: VI

QP Code: R23-CSDLO6013_022023-24

- N.B.** (1) Attempt any three out of the five questions.
(2) Figures to the right indicate full marks.
(3) Assume suitable data if necessary

Marks

- Q.1 (a) A manufacturer of television sets knows that on an average 5% of their product is defective. They sell television sets in consignment of 100 and guarantee that not more than 2 sets will be defective. What is the probability that the TV set will fail to meet the guaranteed quality? *0.001* 05

- (b) Calculate coefficient of correlation for the following data. 05

X	12	9	8	10	11	13	7
Y	14	8	6	9	11	12	3

Highlight the type of correlation between X and Y.

- (c) Give the details of critical region and acceptance region in one tailed and two tailed test of hypothesis with neat diagrams. 05

- (d) In a trip, organized by a college, there were 80 people, each of whom paid Rs. 15.50 on an average. There were 60 students, each of whom paid Rs 16. Members of teaching staff were charged at a higher rate, the number of servants (all males) were 6 and they were not charged anything. The number of females were 20% of the total and there was only one female staff member. Present the following information in a suitable table. 05

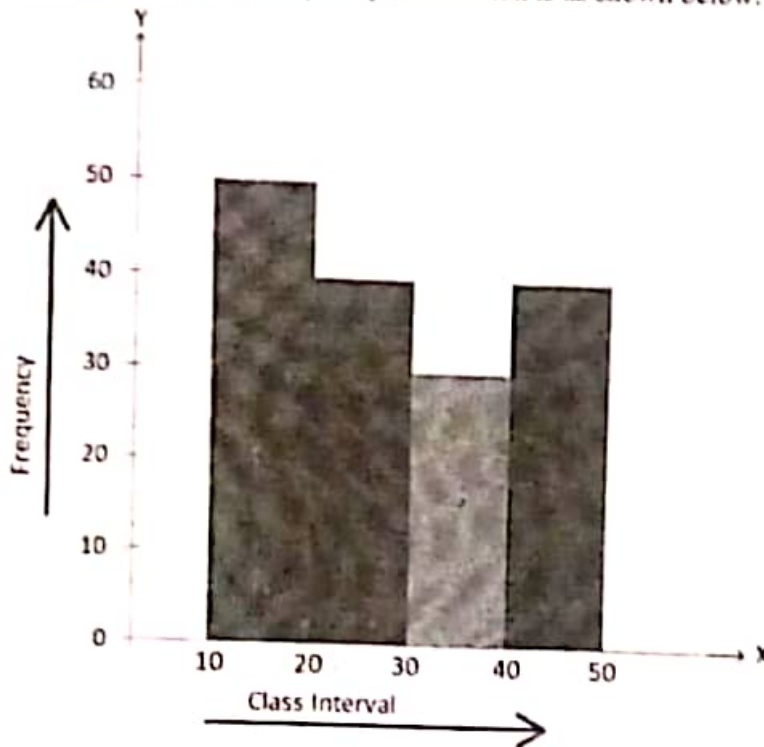
- Q.2 (a) The average test marks in a class is 80. The standard deviation is 6. If the marks are distributed normally, how many students in a class of 200 receive marks between 70 and 90? *0.90308* 05

QP Code: R23-CSDLO6013_022023-24



(b) The histogram for a frequency distribution is as shown below.

05



- Find the class intervals having the greatest and the least frequencies.
- What is the cumulative frequency of the class interval 30 - 40? *116*
- Construct the frequency table of the distribution.

(c) For the given lines of regression $3X - 2Y = 5$ and $X - 4Y = 7$. Find:

10

- mean of X and Y respectively *1.6 0.6*
- regression coefficients *2/3 1/4*
- coefficient of correlation *0.408*

Q.3 (a) What are the various methods of collecting statistical data? Which of these is most reliable and why? *5*

05

(b) In a certain trivariate distribution: $r_{12} = 0.7$, $r_{31} = 0.6$, $r_{23} = 0.6$. Determine the partial regression coefficient $r_{12.3}$

05

(c) In a survey of buying habits, 400 women shoppers are chosen at random in supermarket 'A'. Their average weekly food expenditure is Rs. 250 with standard deviation of Rs. 40. For 400 women shoppers chosen at random in supermarket 'B', the average weekly food expenditure is Rs. 220 with standard

10

QP Code: R23-CSDLO6013_022023-24



1800 219 755
022 261 1849
022 261 1850

deviation of Rs. 55. Test at 1% level of significance whether the average weekly food expenditure of the two populations of shoppers are equal.

- Q.4 (a) Draw all possible samples of size 2 from the population 8, 12, 16 with replacement and without replacement. Considering the samples without replacement, verify that expectation of sample mean is equal to the population mean and find variance of sample mean. 10
- (b) Given the following: $r_{12} = 0.8$, $r_{13} = 0.6$, $r_{23} = 0.5$, $\sigma_1 = 10$, $\sigma_2 = 8$, $\sigma_3 = 5$. Determine the regression equation of:
i. X_1 on X_2 and X_3
ii. X_2 on X_1 and X_3 10

- Q.5 (a) There is a general belief that high income families send their children to public schools and low income families send their children to government schools. To verify this, one thousand families were randomly selected in a city and results obtained on them were as follows: 10

Income Level	Public School	Government School
Low	100	200
High	500	200

Use the Chi-Square test to determine whether the income level and type of schooling are associated.

- (b) A medicine was found to be effective for 9 patients in 8 days on an average with standard deviation of 2.2 days. Another medicine administered to another group of 8 patients was found to be effective in 6 days on an average with standard deviation of 2.6 days. Use 5% level of significance to test the null hypothesis that the two medicines are equally effective. 10

X—X—X—X